

APPENDIX 9 – CONDITION RANGES AND CROSS-TYPE CONNECTIONS

Updating Conditions to Define Their Ranges

The purpose is five-fold. First, to understand the scope and focus of each condition type. Second, how condition categories (that constitute each condition type) operate or manifest. Third, a theorization of why each condition type matters for outcomes. Fourth, the unifying factors for cases representing a particular category. Fifth, a theorization of how condition types interact with each other.

Condition type 1: technology practices (26 rows in ‘master conditions table’ – Appendix 2, 6 categories)

Technology Practices encompasses how initiatives actually build technology: the concrete choices about development processes, participation, and the forms of control embedded in technical architecture.

Technology Practices conditions operate through enabling mechanisms: they "enable," "ground," "design," "steward," and "open" rather than "protect" or "legitimize," distinguishing their functional role from Governance or Politics conditions. Within the Design-Affordance-Sovereignty framework, Technology Practices constitutes the design layer. The choices made here - who participates in development, whether code is open or proprietary, whether architecture enables local control or demands centralized coordination - shape what forms of sovereignty become achievable downstream.

Range analysis

Technology Practices is one of the most prevalent condition types in the dataset, contributing 89 condition-outcome connections (12.7% of total). This prevalence is not accidental: how technology gets built turns out to be central to whether community-led digital stewardship succeeds.

The range spans four dimensions. First, where does intervention occur? Some initiatives focus on governing data after collection; others intervene in the development process itself; still others work at the level of technical architecture. Second, at what scale? Practices range from local community projects to global commons infrastructure. Third, whose knowledge matters? Some initiatives center Western technical expertise while others deliberately embed Indigenous and community epistemologies. Fourth, how open? The spectrum runs from challenging proprietary enclosure to full commons stewardship.

Not all Technology Practices contribute equally to outcomes. Open Development (TP2, resonance=22) and Multi-Scale Architecture (TP3, resonance=24) function as generalist enablers. They connect broadly across outcome types, appearing wherever initiatives achieve democratic governance, create community value, navigate tensions, or generate systemic impact. Culturally-Grounded Development (TP5, resonance=5), by contrast, operates as a specialist condition. It appears in only three cases, but for those cases pursuing cultural and linguistic sovereignty, it proves essential. This pattern suggests that while some technology practices are broadly applicable, others become necessary only for specific sovereignty goals, and that culturally-embedded development may be underutilized relative to its importance.

One finding deserves particular attention: Democratic Data Authority (TP1) shows 75% coverage for Community Data Sovereignty outcomes (DP2). This near-necessary condition relationship means that initiatives rarely achieve data sovereignty without first establishing democratic authority over data governance. The technical and the political cannot be separated.

Condition categories

Democratic Data Authority (TP1) creates mechanisms for communities, citizens, or data subjects to exercise governance over data that was previously controlled by

corporations, states, or researchers. The contexts vary. DECODE and MiData address corporate health data control, FNIGC and MAIAM NAYRI WINGARA confront colonial data exclusion, WAO tackles algorithmic opacity in platform work, but the mechanism is consistent: restructuring who decides how data is collected, used, and shared.

Representative Cases: DECODE, FNIGC, MAIAM NAYRI WINGARA, MiData, OpenSAFELY, WAO.

Open Development and Transparency (TP2) challenges proprietary control through collaborative processes that enable scrutiny, modification, and redistribution. This category ranks among the five most broadly connected conditions in the entire dataset. Linux Foundation and WikiHouse open hardware and software development; Loomio and Matrix open decision-making and communication tools; OpenStreetMap opens geographic knowledge; Sensorica opens research and development itself. The mechanism works by constructing commons through transparent production, removing proprietary gatekeeping so that collective participation becomes possible.

Representative Cases: DECODE, Linux Foundation, Loomio, Matrix, OpenStreetMap, Sensorica, WikiHouse.

Multi-Scale Architecture Design (TP3) deliberately structures technology to work across multiple scales simultaneously. This is a strategic choice with profound governance implications. Guifi.net, Matrix, and OpenAgriNet design federated systems where local nodes maintain autonomy while participating in broader networks. The finding that Multi-Scale Architecture shows 83% coverage for Federated Governance outcomes (BL1) reveals that technical architecture is foundational to how initiatives navigate tensions between local control and trans-local coordination. You cannot achieve federated governance through governance mechanisms alone.

Representative Cases: Australian Research Data Commons, Guifi.net, Matrix, Open Mobility Foundation, OpenAgriNet, SOLshare.

Community-Centered Design Processes (TP4) ensures that those affected by technology have meaningful voice in its creation. Only two cases fall into this category: DAIR centering communities affected by AI bias, The Drivers Cooperative centering driver concerns in platform design. The mechanism is distinct from Democratic Data Authority. Where TP1 addresses governance of existing data, Community-Centered Design intervenes earlier: who participates in creating the technology in the first place. The small number of cases may indicate an underexplored pathway.

Representative Cases: DAIR, The Drivers Cooperative.

Culturally-Grounded Development (TP5) roots technology in specific cultural contexts and knowledge systems rather than importing generic solutions. Masakhane builds African language AI through community linguistic knowledge; Monlam AI develops Tibetan language technology amid political suppression; Te Hiku Media creates Māori language AI for cultural revitalization. This category's low prevalence but high concentration in cultural sovereignty outcomes (67% coverage for IM3) suggests a tension worth noting: if meaningful sovereignty requires a reflection of community epistemologies, more initiatives may need to adopt culturally-grounded practices than currently do.

Representative Cases: Masakhane, Monlam AI, Te Hiku Media.

Commons-Based Resource Stewardship (TP6) governs technology as shared resources managed collectively. OpenStreetMap maintains mapping data as global commons; Sensorica stewards open research infrastructure; Guifi.net manages telecommunications as community resource. The mechanism differs from Open Development in emphasis: where TP2 focuses on the production process, Commons Stewardship focuses on ongoing governance of the resource produced.

Representative Cases: Guifi.net, OpenStreetMap, Sensorica.

Cross-Type connections

Technology Practices does not operate in isolation. Its relationship with Governance is bidirectional and often definitional: Governance conditions structure authority, while Technology Practices conditions enable the technical affordances through which that authority operates. Decentralized governance requires both: authority distribution (GV1) and technical architecture that supports distributed decision-making.

The connection to Resources is equally significant. Community Capacity Development (RS1) frequently co-occurs with Technology Practices because building technical capacity within communities enables adoption and adaptation of these practices. Without capacity, technology practices remain aspirational.

Purpose conditions orient which technology practices initiatives prioritize. Cultural Preservation Purpose (PU2) shows deep co-occurrence with Culturally-Grounded Development (TP5) for cultural sovereignty outcomes; the value commitment drives the technical approach. Similarly, the relationship with Partners shapes what technology practices are feasible: Open Development attracts collaboration, while Multi-Network Positioning (PT2) creates the relational context within which technology practices can spread and scale.

Condition type 2: resources (23 rows, 6 categories)

Resources covers how initiatives mobilize, allocate, and transform economic and human capital to build sovereignty capacity. Unlike Technology Practices, which shapes what is designed, Resources addresses what enables design to happen at all. These are the material and capability foundations without which architectural choices remain aspirational. Resources conditions operate through constructive mechanisms: they "build," "support," "align," "experiment," and "sustain" rather than "structure" or "enable." This distinction

matters: where Technology Practices creates affordances, Resources constructs the capacity to act on them.

Range analysis

With 91 condition-outcome connections (13.0% of total), Resources confirms what practitioners know intuitively: you cannot build alternative technology without material foundations. The range spans from investments in internal community capabilities to restructuring external economic flows.

Four dimensions capture this variation. First, level of intervention: some conditions develop individual skills (RS2) while others restructure collective capacity (RS1) or transform entire value systems (RS3). Second, direction of flow: some conditions address resource acquisition (RS4, RS5) while others address retention and circulation (RS3). Third, source: funding may come from reciprocal exchanges (RS5), alternative economic models (RS4), or capacity investments (RS1). Fourth, temporality: some resources enable immediate action; others build long-term sustainability.

The quantitative analysis shows how Community Capacity Development (RS1, resonance=29) ranks second among all 56 condition categories. This broad connectivity reflects a theoretical point the D-A-S framework emphasizes: sovereignty requires actual capability, not just formal authority or good design. The near-tautological finding that RS1 shows 100% coverage for Community Capacity Building outcomes (IM2) confirms that explicit investment in community capabilities translates directly to impact, a validation of intentional capacity-building strategies.

Economic Value Redistribution (RS3) shows a different pattern: 100% coverage for Economic Value Transformation outcomes (IM4). When initiatives successfully transform economic value flows, they invariably do so through explicit redistribution mechanisms. There is no backdoor to economic transformation.

Condition categories

Community Capacity Development (RS1) invests in building community-level capabilities to exercise agency over technology and governance. The mechanism is developing internal human capital (skills, knowledge, confidence) that enables communities to participate meaningfully rather than depend on external actors. DAIR builds AI research capacity in affected communities; FNIGC develops First Nations data governance capabilities; Masakhane creates distributed African AI research capacity; Slum Dwellers International enables communities to generate their own data. The prevalence of this condition type across the dataset, and its exceptional resonance with outcomes, confirms what decolonial theorists argue: meaningful self-determination requires communities to possess the internal capabilities to exercise governance.

Representative Cases: DAIR, FNIGC, Masakhane, Participatory Brazil, Slum Dwellers International, Stop LAPD Spying Coalition.

Access and Skills Democratization (RS2) reduces barriers to individual participation through training, equipment access, and technological empowerment. While RS1 builds collective capacity, RS2 focuses on individual-level enablement, giving people the tools and skills to participate in digital and economic systems from which they were excluded. CLEAR democratizes access to scientific equipment; Detroit CTC addresses digital skills gaps; Digital Empowerment Foundation supports micro-entrepreneurs; Zenzeleni builds technical self-reliance.

Representative Cases: CLEAR, Detroit Community Technology Project, Digital Empowerment Foundation, Zenzeleni.

Economic Value Redistribution (RS3) restructures value flows to redirect economic benefits from external extraction to community retention. This is not about generating more value but about changing who captures it. CoopCycle and The Drivers Cooperative redirect

value from platform shareholders to workers; Guifi.net keeps telecommunications revenues circulating locally; Namma Yatri eliminates the 40% commission extraction typical of ride-hailing platforms. The 100% coverage with Economic Value Transformation outcomes indicates this condition is necessary.

Representative Cases: CoopCycle, Fairbnb.coop, Guifi.net, Namma Yatri, Solar Commons, The Drivers Cooperative.

Alternative Economic Experimentation (RS4) creates new economic models beyond conventional market or state provision. Where RS3 redirects value within existing flows, RS4 generates entirely new circulation logics. Sensorica's open value network, Enspiral's collaborative funding, and Research Data Alliance's shared infrastructure represent experiments in alternative economic organization.

Representative Cases: Enspiral, Research Data Alliance, Sensorica.

Partnership-Based Resource Exchange (RS5) generates resources through reciprocal relationships rather than market transactions or philanthropic dependency. These conditions create sustainable resource flows embedded in relationships of mutual benefit.

Representative Cases: Australian Research Data Commons, Linux Foundation, OpenAgriNet.

Sustainable Funding Innovation (RS6) addresses the perpetual challenge of financial sustainability through alternative funding models that don't compromise mission.

Representative Cases: Groupmuse, Som Energia, WikiHouse.

Cross-Type connections

Resources conditions interact most directly with Governance. Ownership-based democratic control (GV2) frequently requires economic restructuring (RS3) to be meaningful, and decentralized authority (GV1) often needs capacity development (RS1) to be exercised.

The relationship with Technology Practices is foundational: technology practices represent design choices, but implementing those choices requires resource investments. Community Capacity Development (RS1) frequently co-occurs with Technology Practices conditions because building capacity enables adoption of new technical approaches.

Purpose conditions orient resource mobilization strategies. Initiatives with cultural preservation purposes (PU2) prioritize cultural capacity, while those with ecological transformation purposes (PU5) prioritize sustainability-oriented resources.

Condition type 3: governance (22 rows, 6 categories)

Governance conditions refer to how initiatives structure authority, make decisions, and coordinate collective action. Where Resources provides the material foundation and Technology Practices shapes technical affordances, Governance determines who holds decision-making power and how it flows. Governance conditions operate through structural mechanisms: they "structure," "distribute," "govern," "integrate," and "externalize." These verbs reflect that governance is fundamentally about configuring authority; not just processes but the architecture of power itself.

Range analysis

Governance spans 79 condition-outcome connections (11.3% of total). The range addresses a fundamental tension: how can organizations remain democratic as they grow, coordinate across scales, integrate specialized expertise, and engage external institutions?

The categories reveal multiple solutions to this tension. Some initiatives distribute authority that was previously concentrated (GV1). Others embed democratic control in ownership structure (GV2). Still others create federated architectures that enable local autonomy within broader coordination (GV3). The diversity reflects that there is no single path to democratic governance, as different contexts demand different structural solutions.

Decentralized Authority Distribution (GV1) shows particularly strong connections to Decentralized Organizational Governance outcomes (DP1). The relationship is definitional: "you cannot have decentralized governance without decentralized authority distribution." This may seem tautological, but it carries an important implication, decentralized governance requires intentional authority restructuring beyond good intentions or participatory rhetoric.

Federated Multi-Scale Coordination (GV3) shows different patterns, connecting strongly to Balance outcomes. This makes sense theoretically: federation is precisely a governance response to scalar tensions, enabling initiatives to maintain local autonomy while participating in trans-local coordination.

Condition categories

Decentralized Authority Distribution (GV1) addresses hierarchical concentration by distributing decision-making power to affected parties. The challenges span corporate hierarchies (DAIR), project management (Detroit CTC), organizational control (Enspiral, Loomio), technology development (Nea Guinea/Tzoumakers), and state power (Participatory Brazil). Despite different contexts, the mechanism is similar: redistributing authority that was concentrated. This category directly addresses the dissertation's first research question concerning democratic practices. Decentralization is more than a value preference, and there is strong support for the notion that it is a structural condition for democracy.

Representative Cases: DAIR, Detroit Community Technology Project, Enspiral, Loomio, Nea Guinea/Tzoumakers, Participatory Brazil.

Ownership-Based Democratic Control (GV2) uses ownership structure as the lever for governance transformation. The insight is straightforward: ownership determines governance outcomes. Worker ownership (CoopCycle, The Drivers Cooperative) and community ownership (Groupmuse, MiData, Som Energia) both embed democratic control in organizational form. This category shows strong connections to Community Wealth

Retention outcomes (CV3), confirming that ownership structure shapes both governance and value distribution.

Representative Cases: CoopCycle, Groupmuse, MiData, Som Energia, The Drivers Cooperative.

Federated Multi-Scale Coordination (GV3) creates governance architectures that resolve the tension between local autonomy and network coordination. Guifi.net and Matrix employ federated models where local nodes maintain self-governance while participating in broader networks; MiData federates health data cooperatives across jurisdictions. The 83% coverage with Federated Governance outcomes (BL1) confirms that federation is the primary governance pathway to navigating scalar tensions.

Representative Cases: Guifi.net, Matrix, MiData.

Democratic Scaling Strategies (GV4) explicitly addresses how to maintain democratic character as organizations grow. Scale creates pressure toward hierarchy and exclusion; this category represents intentional resistance to those pressures. CoopCycle navigates scaling within its federation; FNIGC navigates engagement across scales from national to global.

Representative Cases: CoopCycle, FNIGC.

Inclusive Expertise Integration (GV5) develops mechanisms to integrate specialized knowledge with broad participation. This addresses a governance challenge distinct from decentralization: how can technical expertise be integrated without subordinating democratic authority? Sensorica's stigmergic coordination and WAO's inclusive expertise model represent different solutions.

Representative Cases: Sensorica, Worker Algorithm Observatory.

Democratization of External Processes (GV6) creates new participation channels in external governance structures (state planning, institutional policy-making) that were

previously technocratic or exclusionary. Unlike GV1-GV5, which address internal organizational governance, GV6 focuses on how communities engage with external institutions. Participatory Brazil democratizes government policy development; Slum Dwellers International democratizes urban planning. Both recognize that internal democracy is insufficient if external governance remains exclusionary.

Representative Cases: Participatory Brazil, Slum Dwellers International.

Cross-Type connections

Governance connects bidirectionally with Technology Practices. GV3 (federated governance) frequently co-occurs with TP3 (multi-scale architecture) because scalar governance solutions require technical architecture that supports them.

The connection with Resources is equally important: GV2 (ownership-based control) enables RS3 (value redistribution) by ensuring that those who create value have governance authority over it.

Purpose conditions shape which governance approaches initiatives prioritize. Indigenous epistemological centering (PU1) leads to governance structures that reflect Indigenous protocols, while resistance-construction integration (PU4) leads to governance that balances oppositional and generative functions.

Condition type 4: technology, tools, data, products, and customers (22 rows, 7 categories)

Technology Tools includes the specific technological artifacts, infrastructures, and products that initiatives develop or control, referring to the material layer of sovereignty. Where Technology Practices addresses how technology is built, Technology Tools addresses what gets built and deployed. These conditions operate through operationalizing mechanisms: they "facilitate," "instantiate," "operationalize," "ground," and "enable." The distinction from Technology Practices is significant: practices shape processes and participation; tools represent the material outputs through which sovereignty is exercised.

Range analysis

With 78 condition-outcome connections (11.1% of total), Technology Tools represents the tangible manifestation of community-led digital stewardship. The range spans from physical infrastructure (telecommunications networks, energy grids) to digital platforms (decision-making tools, mapping systems) to data resources (Indigenous language corpora, health records). These artifacts embody sovereignty. Community-owned infrastructure creates different affordances than corporate-owned infrastructure, regardless of governance intentions.

The quantitative analysis unearthed a striking necessary condition relationship: Physical Infrastructure (TT1) shows 100% coverage for Community-Controlled Infrastructure outcomes (DP3). One cannot achieve infrastructure sovereignty through governance declarations alone, as material control over physical or digital systems is necessary.

Condition categories

Community-Controlled Physical Infrastructure (TT1) builds and operates essential infrastructure under community ownership, particularly where commercial providers are absent or extractive. Detroit CTC builds community-owned digital networks; Guifi.net operates community telecommunications across rural Spain; SOLshare creates peer-to-peer solar microgrids; TIC A.C. and Zenzeleni provide telecommunications to Indigenous and rural communities respectively. The 100% coverage with Infrastructure Control outcomes indicates a necessary relationship. There is no governance-only path to controlling essential services.

Representative Cases: Detroit Community Technology Project, Guifi.net, SOLshare, Telecomunicaciones Indígenas Comunitarias, Zenzeleni.

Culturally-Grounded Technology Artifacts (TT2) develops technology specifically adapted to cultural and linguistic contexts that mainstream technology ignores. Digital Empowerment Foundation creates tools for Indian local languages; Monlam AI develops technology embedded in Buddhist scholarly traditions; Te Hiku Media builds Māori language AI that mainstream providers cannot serve. The mechanism is embedding cultural specificity into technological form, creating tools that reflect community realities.

Representative Cases: Digital Empowerment Foundation, Monlam AI, Nea Guinea/Tzoumakers, Te Hiku Media.

Indigenous Data Sovereignty Infrastructure (TT3) develops purpose-built tools for Indigenous data governance. FNIGC builds First Nations data infrastructure; GIDA creates metadata systems that embed Indigenous governance protocols. This category operationalizes sovereignty-as-capacity: sovereignty over data requires actual technical infrastructure for Indigenous control in addition to recognition within existing systems.

Representative Cases: FNIGC, GIDA.

Participatory Design Platforms (TT4) builds technology that facilitates transparent, participatory decision-making. Loomio creates asynchronous deliberation tools; Open Mobility Foundation develops public-facing technical governance; Participatory Brazil enables citizen participation in government. The technology itself embodies participatory values, creating affordances for democratic engagement.

Representative Cases: Loomio, Open Mobility Foundation, Participatory Brazil.

Digital Commons Resources (TT5) creates and maintains digital resources as collectively governed commons. OpenStreetMap maintains mapping data as global commons; P2P Foundation documents commons-based knowledge; Research Data Alliance develops shared research infrastructure.

Representative Cases: OpenStreetMap, P2P Foundation, Research Data Alliance.

Open Communication and Collaboration Systems (TT6) provides transparent, interoperable systems for communication and coordination. Matrix develops open secure communications; Linux Foundation maintains essential open-source software; Sensorica creates open collaboration infrastructure.

Representative Cases: Linux Foundation, Matrix, Sensorica.

Decentralized Technical Architecture (TT7) implements technically decentralized systems where no single point of control exists. SOLshare's peer-to-peer energy trading represents this specialized category. Low resonance reflects limited replication, though the theoretical significance — infrastructure that structurally prevents centralization — is substantial.

Representative Cases: SOLshare.

Cross-Type connections

Technology Tools connects most directly to Technology Practices. TT1 (physical infrastructure) often results from TP3 (multi-scale architecture design), and TT2 (culturally-grounded artifacts) emerges from TP5 (culturally-grounded development).

The relationship with Governance is reciprocal: governance choices shape what tools get built, but tool characteristics also constrain governance possibilities. Decentralized infrastructure (TT7) enables but does not guarantee decentralized governance.

Resources conditions are often prerequisites: building and operating infrastructure requires sustained resource investment, making RS1 (capacity development) and RS3 (value redistribution) frequent co-occurrences.

Condition type 5: partners (20 rows, 5 categories)

Partners addresses how initiatives build, maintain, and navigate external relationships, or the relational dimension of sovereignty. Sovereignty emerges through strategic positioning within networks of allies, funders, collaborators, and sometimes adversaries. Partners

conditions operate through connective mechanisms: they "exchange," "connect," "coordinate," "network," and "translate." These verbs reflect that partnership is fundamentally about creating relational pathways for resource and knowledge flows while protecting community autonomy.

Range analysis

Partners spans 76 condition-outcome connections (10.9% of total). The range is underscored by a central tension: initiatives need external relationships to access resources, build collective power, and achieve scale, but those relationships create risks of co-optation, knowledge extraction, and sovereignty erosion.

The categories represent different navigational strategies. Some conditions focus on building beneficial relationships (PT1, PT2). Others focus on protecting sovereignty within relationships (PT3, PT4). The quantitative analysis shows that Multi-Network Strategic Positioning (PT2, resonance=24) ranks among the most broadly connected conditions in the dataset, confirming that strategic network positioning is not optional for successful initiatives.

Condition categories

Reciprocal Knowledge Exchange (PT1) builds partnerships structured around mutual benefit. ARDC develops bidirectional research partnerships; CLEAR centers community knowledge alongside scientific expertise; Digital Empowerment Foundation co-develops with communities; MAIAM NAYRI WINGARA ensures research benefits Aboriginal communities. The mechanism is reciprocity: partnerships where value flows in both directions.

Representative Cases: Australian Research Data Commons, CLEAR, Digital Empowerment Foundation, MAIAM NAYRI WINGARA.

Multi-Network Strategic Positioning (PT2) builds presence and influence across multiple network types: movement networks, policy networks, funding networks, and peer networks. P2P Foundation positions itself across commons initiatives globally; Slum Dwellers International connects slum communities across countries; Te Mana Raraunga bridges Indigenous data sovereignty movements. It overcomes isolation and enables collective action. High resonance reflects that multi-network positioning creates affordances across outcome types: resource mobilization (Community Value), legitimacy (Democratic Practices), scaling (Impact), and navigation (Balance).

Representative Cases: Enspiral, GIDA, Open Mobility Foundation, P2P Foundation, Slum Dwellers International, Te Mana Raraunga.

Community Sovereignty Protection in Partnerships (PT3) navigates external relationships while maintaining community control. Fairbnb.coop balances platform expansion with local accountability; Monlam AI maintains monastic oversight despite technical partnerships; Te Hiku Media protects Māori data control while exploring collaborations. The mechanism is boundary-setting that enables collaboration without sovereignty erosion. This is a recognition that partnerships always carry risks.

Representative Cases: Fairbnb.coop, Monlam AI, Te Hiku Media.

Expertise-Community Knowledge Integration (PT4) develops hybrid governance that integrates specialized expertise with community knowledge without subordinating community authority. CLEAR navigates scientific-community knowledge tensions; Linux Foundation balances corporate funding with community governance; Masakhane integrates AI expertise with linguistic community knowledge. The aim is to achieve partnerships where different knowledge forms are combined instead of subsuming one another.

Representative Cases: CLEAR, Linux Foundation, Masakhane, OpenSAFELY.

Cross-Sector Collaboration Networks (PT5) bridges typically siloed sectors (corporate, academic, community) to enable novel collaboration forms. Sensorica's open value network represents this specialized category. Low resonance reflects limited replication, though the model addresses real barriers to cross-sector innovation.

Representative Cases: Sensorica.

Cross-Type connections

Partners intersects with Politics conditions around external system engagement. PT2 (multi-network positioning) enables PO1 (policy advocacy) by creating the relational infrastructure for collective voice.

The connection to Purpose is significant: Indigenous epistemological centering (PU1) shapes how partnerships are structured (PT1 reciprocity, PT3 sovereignty protection), while resistance-construction integration (PU4) shapes which networks initiatives engage.

Resources conditions frequently co-occur because partnerships are a primary mechanism for resource acquisition (RS5 partnership-based resource exchange).

Condition type 6: purpose (16 rows, 6 categories)

Purpose captures the foundational orientations, values, and worldviews guiding initiatives, or the "why" behind technological and organizational choices. Unlike other condition types that address what initiatives do or how they organize, Purpose addresses what they believe worth pursuing and how they define success. Purpose conditions operate through orienting mechanisms: they "orient," "ground," "motivate," "direct," and "obligate."

Range analysis

With 59 condition-outcome connections (8.4% of total), Purpose represents the normative foundation on which other conditions rest. The range indicates that different value systems produce different design choices.

The quantitative analysis reveals important specialist patterns. Indigenous Epistemological Centering (PU1) shows 100% coverage for Cultural Governance Integration outcomes (DP6). Cultural Preservation Purpose (PU2) also shows 100% coverage for Cultural and Linguistic Sovereignty outcomes (IM3); linguistic preservation does not happen accidentally but requires explicit purpose orientation. These necessary condition relationships indicate that purpose is both motivational and structurally determinative of what outcomes become achievable. An initiative's purpose shapes which technology practices it prioritizes, which governance structures it builds, which partnerships it pursues, and how it defines success.

Condition categories

Indigenous Epistemological Centering (PU1) makes Indigenous knowledge systems foundational to organizational purpose; the epistemological ground from which decisions flow. FNIGC, GIDA, Te Hiku Media, and Te Mana Raraunga all operate from Indigenous worldviews as primary. The challenges addressed involve colonial erasure and Western governance disconnection. The mechanism is epistemic recentering: making what was marginalized foundational.

Representative Cases: FNIGC, GIDA, Te Hiku Media, Te Mana Raraunga.

Cultural Preservation and Linguistic Sovereignty (PU2) frames organizational purpose as cultural and linguistic survival. For Masakhane, Monlam AI, and Te Hiku Media, technology serves cultural preservation instead of efficiency or profit. The challenges are existential: language exclusion threatens cultural survival.

Representative Cases: Masakhane, Monlam AI, Te Hiku Media.

Value System Reconceptualization (PU3) challenges narrow economic definitions of value. Groupmuse prioritizes cultural over commercial value; MAIAM NAYRI WINGARA challenges data-as-property frameworks; Solar Commons challenges narrow

sustainability metrics. The mechanism is redefining what counts, or changing the value framework. This directly addresses the dissertation's second research question by suggesting that community wealth requires reconceptualized definitions of wealth.

Representative Cases: Groupmuse, MAIAM NAYRI WINGARA, Solar Commons.

Resistance-Construction Integration (PU4) combines critical resistance with constructive alternative-building as unified purpose. TIC A.C. resists regulatory burdens while building community networks; WAO resists algorithmic opacity while building worker knowledge. The mechanism refuses separation between critique and creation, viewing opposition and alternatives as two faces of the same project.

Representative Cases: TIC A.C., Worker Algorithm Observatory.

Ecological and Social Transformation (PU5) grounds purpose in broader transformation visions: ecological sustainability, feminist economics, intersectional justice. Les Mercedes pursues feminist economic transformation; Som Energia embeds energy transition in social transformation. Technology serves these broader purposes first before other objectives.

Representative Cases: Les Mercedes, Som Energia.

Democratic Citizenship Obligation (PU6) frames participation as civic responsibility. Participatory Brazil treats citizen participation as obligation; TIC A.C. prioritizes local needs over external agendas. The mechanism orients toward citizenship and community priority over service provision or market competition.

Representative Cases: Participatory Brazil, TIC A.C.

Cross-Type connections

Purpose conditions shape all other condition types. PU1 (Indigenous epistemological centering) determines which governance structures (GV) and technology practices (TP) are

appropriate. PU2 (cultural preservation) drives TP5 (culturally-grounded development) and TT2 (culturally-grounded artifacts); the value commitment shapes technical choices.

PU4 (resistance-construction integration) influences Politics conditions, shaping whether initiatives engage through policy influence (PO1), resistance (PO2), or strategic navigation (PO3).

Condition type 7: organizational information (15 rows, 6 categories)

Org Info captures organizational identity, meaning how initiatives define their organizational selves. This is not about what organizations do but what they are. Org Info conditions operate through positioning mechanisms: they "define," "position," "embody," "frame," and "characterize." These conditions recognize that organizational form shapes possibilities for action: being a cooperative creates different possibilities than being a foundation; positioning as municipal infrastructure creates different affordances than positioning as movement organization.

Range analysis

Org Info spans 58 condition-outcome connections (8.3% of total). The range addresses multiple dimensions of organizational identity: ownership structure, governance architecture, value circulation patterns, production mode, scalar positioning, and public accountability. Most organizations combine multiple dimensions; Loomio is both worker-owned (OI1) and commons-based (OI4); Guifi.net is both decentralized (OI2) and anti-extractive (OI3).

The quantitative analysis reflects the relative scarcity of public / municipal models compared to community-based and cooperative forms, which may signal an underexplored organizational pathway.

Condition categories

Worker / Community Ownership Models (OI1) defines organizations through ownership structure. Loomio is collectively owned; MiData is a data cooperative; Namma Yatri is worker-owned. The mechanism is identity through ownership. The organizational form itself is the alternative to extractive models, not policies that can be reversed but structure that persists.

Representative Cases: Loomio, MiData, Namma Yatri.

Decentralized Governance Architectures (OI2) defines organizations through how decisions are made. Guifi.net's identity centers on decentralization; Participatory Brazil's identity centers on hybrid democratic architecture. The mechanism is identity through governance form. The organization IS its decision-making structure.

Representative Cases: Guifi.net, Participatory Brazil.

Anti-Extractive Value Circulation (OI3) defines organizations through how value flows. Fairbnb.coop, Som Energia, and The Drivers Cooperative all define themselves against extraction; their identity is their value circulation logic.

Representative Cases: Fairbnb.coop, Som Energia, The Drivers Cooperative.

Commons-Based Production Identity (OI4) defines organizations through production mode. Matrix produces open protocols; P2P Foundation produces commons-based knowledge; WikiHouse produces open-source housing designs. The mechanism is identity through how outputs are created and governed.

Representative Cases: Matrix, P2P Foundation, WikiHouse.

Local-Global Scalar Positioning (OI5) defines organizations through their navigation of scalar tensions. Digital Empowerment Foundation addresses local capacity within global digital divides; Les Mercedes and Slum Dwellers International prioritize local needs while engaging globally. The mechanism is identity through scalar positioning.

Representative Cases: Digital Empowerment Foundation, Les Mercedes, Slum Dwellers International.

Municipal / Public Digital Governance (OI6) positions organizations as public infrastructure governance bodies. Open Mobility Foundation defines itself through municipal accountability. This unique positioning reflects a distinct organizational model: public digital infrastructure governance.

Representative Cases: Open Mobility Foundation.

Cross-Type connections

Org Info conditions provide the organizational bedrock for other conditions to operate. OI1 (ownership models) enables GV2 (ownership-based democratic control); OI3 (anti-extractive circulation) operationalizes RS3 (value redistribution); OI4 (commons-based production) supports TP6 (commons stewardship).

The relationship with Purpose is foundational: purpose orientations (PU) shape organizational identity choices (OI), which in turn constrain and enable governance (GV), technology (TP), and resource (RS) conditions.

Condition type 8: politics (13 rows, 3 categories)¹

Politics addresses how initiatives engage with political systems, policy landscapes, and power structures beyond organizational boundaries. Community-led alternatives cannot succeed if broader political-economic contexts remain hostile. Politics conditions represent strategies for reshaping that context. These conditions operate through legitimizing mechanisms: they "advocate," "legitimize," "drive," "challenge," and "integrate." The dominant verb pattern reflects that community-led initiatives operate in contested legitimacy environments where they must actively construct their right to exist and govern.

¹ The conditions up to this point were internal conditions, describing organizational dynamics. Politics and the remaining three condition types (regulation, social context, problem) provide this dissertation's external conditions.

Range analysis

Politics contribute 58 condition-outcome connections (8.3% of total). Policy and Systemic Transformation Advocacy (PO1, resonance=41) is the single most broadly connected condition category across the entire dataset. Nothing else comes close.

This finding demands attention. It means that regardless of how well initiatives structure governance, build capacity, or design technology, engagement with external political-economic systems is foundational to achieving outcomes and impact. PO1 "directly produces" policy change while also "legitimizing" other practices. The same condition operates through multiple mechanisms depending on outcome context.

Condition categories

Policy and Systemic Transformation Advocacy (PO1) works to change regulatory environments, policy frameworks, and systemic structures. This is the dominant form of political engagement. CLEAR influences environmental governance; FNIGC, GIDA, and Te Mana Raraunga advance Indigenous data sovereignty recognition; Matrix shapes communications standards; P2P Foundation brings commons frameworks into policy discourse; Stop LAPD Spying Coalition defeats surveillance programs; The Drivers Cooperative informs platform regulation. The mechanism is reshaping the political-economic environment to be more supportive of community-led alternatives.

Representative Cases: CLEAR, FNIGC, GIDA, Matrix, Open Mobility Foundation, P2P Foundation, Stop LAPD Spying Coalition, Te Mana Raraunga, The Drivers Cooperative.

Resistance-Opposition Integration (PO2) combines critical resistance to harmful systems with constructive alternatives. CLEAR resists expert-community disconnection while building alternative research models; Stop LAPD Spying Coalition resists surveillance expansion while building community knowledge. This differs from PO1's policy influence in its oppositional framing; not reforming systems but fighting them while building alternatives.

Representative Cases: CLEAR, Stop LAPD Spying Coalition.

Strategic Institutional Navigation (PO3) works within hostile or constraining institutional contexts while maintaining transformative commitments. FNIGC engages colonial institutions strategically to advance Indigenous sovereignty; Namma Yatri operates in markets while maintaining non-extractive values. The mechanism is strategic positioning that enables survival and impact within systems that are not supportive.

Representative Cases: FNIGC, Namma Yatri.

Cross-Type connections

Politics conditions connect to Partners through relational infrastructure. PT2 (multi-network positioning) enables PO1 (policy advocacy) by creating the network relationships necessary for collective voice.

The relationship with Purpose shapes political strategy: PU4 (resistance-construction integration) leads to PO2 (resistance-opposition), while PU1 (Indigenous epistemological centering) shapes how initiatives engage with colonial institutions (PO3).

Politics conditions enable outcomes from other domains: policy change (PO1) creates regulatory space for technology practices (TP), governance innovations (GV), and resource redistribution (RS) to succeed.

Condition type 9: regulation (11 rows, 4 categories)

Regulation addresses how initiatives navigate, reshape, or create legal and regulatory frameworks. Legal structures are simultaneously constraints to be navigated and tools for community protection. Regulation conditions operate through protective mechanisms: they "formalize," "assert," "protect," "institutionalize," and "navigate." These verbs distinguish Regulation from Politics: where Politics engages power structures broadly, Regulation engages specifically through legal and institutional frameworks.

Range analysis

Regulation covers 44 condition-outcome connections (6.3% of total). The range reveals a spectrum of regulatory engagement: from working within existing frameworks (RG2, RG3) to creating new legal mechanisms (RG1) to asserting sovereignty beyond state frameworks entirely (RG4).

Most initiatives engage with rather than bypass regulatory systems. This may reflect practical necessity, but it also suggests that sovereignty assertion outside state frameworks remains a frontier requiring further development.

Legal Innovation for Sovereignty Protection (RG1) is the largest category, signaling that appropriation prevention is a central regulatory concern. Te Hiku Media's Kaitiakitanga License, protecting Indigenous language data from open-licensing appropriation, exemplifies the mechanism: creating legal instruments that close pathways for community resource capture.

Condition categories

Legal Innovation for Sovereignty Protection (RG1) creates novel legal frameworks to protect community resources from appropriation. CoopCycle's Coopcycle License prevents corporate capture of cooperative technology; Guifi.net's community network license protects infrastructure from privatization; Solar Commons develops community energy benefit structures; Te Hiku Media's Kaitiakitanga License prevents appropriation of Māori language data. All address appropriation risk through legal innovation that closes capture pathways.

Representative Cases: CoopCycle, Guifi.net, Solar Commons, Te Hiku Media.

Ethical Governance Framework Development (RG2) creates principled frameworks for navigating data governance tensions. Australian Research Data Commons

balances openness with protection of sensitive information; OpenSAFELY balances international standards with NHS-specific requirements.

Representative Cases: Australian Research Data Commons, OpenSAFELY.

Strategic Regulatory Navigation (RG3) develops strategies for operating within frameworks designed for different actors. FNIGC navigates colonial regulatory environments privileging state control; Open Mobility Foundation navigates platform-dominated contexts; TIC A.C. navigates commercial-scale fee structures (\$50,000 annual spectrum fees) never designed for community organizations. The mechanism is strategic adaptation when regulatory reform is not immediately achievable.

Representative Cases: FNIGC, Open Mobility Foundation, TIC A.C.

Self-Determined Sovereignty Assertion (RG4) asserts governance authority beyond state-centric frameworks. GIDA reimagines sovereignty outside state recognition; OpenAgriNet addresses farmer governance needs that global standards ignore. The mechanism is building sovereignty capacity through practice.

Representative Cases: GIDA, OpenAgriNet.

Cross-Type connections

Regulation connects to Politics around external system engagement. RG1 (legal innovation) often follows or supports PO1 (policy advocacy), creating legal structures to protect gains achieved through political engagement.

The relationship with Technology Tools is protective: RG1 innovations often protect TT1 (physical infrastructure) and TT5 (digital commons) from appropriation.

Purpose conditions shape regulatory strategy: Indigenous epistemological centering (PU1) leads to RG4 (self-determined sovereignty) rather than seeking recognition within colonial frameworks.

Condition type 10: social context (8 rows, 3 categories)

Social Context captures the broader social environments within which initiatives operate; conditions that shape what is possible but that initiatives did not create and cannot fully control. Unlike most condition types that represent initiative choices, Social Context represents the external social landscape. These conditions operate through embedding mechanisms: they "embed," "respond," "enable," "situate," and "contextualize." The mechanism vocabulary reflects that social context positions initiatives within broader dynamics.

Range analysis

Social Context has only 35 condition-outcome connections (5.0% of total). This limited explicit coding is itself revealing: social context factors are often implicit in other condition types instead of being independently named. When initiatives describe their challenges and responses, they typically foreground their own choices and not the contexts constraining those choices.

The three categories represent distinct contextual dynamics: inclusion structures that determine who participates (SC1), movement embeddedness that connects initiatives to broader social forces (SC2), and climate crisis as an external condition requiring responses (SC3). Each positions initiatives differently relative to their environments.

Condition categories

Inclusive Participation Structures (SC1) addresses exclusion from participation and decision-making, particularly for marginalized communities. CLEAR addresses Western academic hierarchies marginalizing diverse perspectives; Digital Empowerment Foundation addresses systemic marginalization from digital processes; Participatory Brazil addresses government exclusion of citizen perspectives; Stop LAPD Spying Coalition addresses

surveillance activism led by those distant from impacts. The mechanism is inclusive design that ensures affected communities participate in processes affecting them.

Representative Cases: CLEAR, Digital Empowerment Foundation, Participatory Brazil, Stop LAPD Spying Coalition.

Movement and Network Embeddedness (SC2) explicitly connects initiatives to broader social movements, building power through collective action. GIDA connects to global Indigenous sovereignty movements; OpenStreetMap navigates global-local network dynamics.

Representative Cases: GIDA, OpenStreetMap.

Climate Crisis and Ecological Response (SC3) operates within climate contexts where technological interventions address compounded vulnerabilities. Slum Dwellers International addresses climate vulnerability compounded by informal settlement marginalization; SOLshare addresses climate vulnerability compounded by centralized energy infrastructure failures. The mechanism is technology development that enables climate action integrated with justice.

Representative Cases: Slum Dwellers International, SOLshare.

Cross-Type connections

Social Context conditions represent the external ground within which other conditions operate. SC1 (inclusion structures) shapes what Governance approaches (GV) are necessary; contexts of marginalization require GV1 (decentralized authority) and GV6 (democratization of external processes).

SC2 (movement embeddedness) enables PT2 (multi-network positioning) by providing the movement networks to position within. SC3 (climate response) shapes Purpose (PU5 ecological transformation) and Technology Tools (TT1 infrastructure that addresses climate vulnerability).

Condition type 11: problem (8 rows, 4 categories)

Problem is a condition type that is a little different from the others. It captures the challenges motivating initiatives, the underlying issues that design choices aim to address. Problem conditions represent the "problem space" within which sovereignty is constructed. These conditions "identify," "frame," "diagnose," and "respond to" the challenges that make community-led stewardship necessary in the first place. Unlike Social Context (which describes environments), Problem names specific challenges requiring intervention.

Range analysis

Problem has only 33 condition-outcome connections (4.7% of total). The limited explicit coding suggests that problem identification is often implicit in other condition types as opposed to being independently named. Technology Practices conditions imply problems (proprietary enclosure, data extraction); Governance conditions imply problems (hierarchical concentration, exclusion); Purpose conditions imply problems (colonial erasure, cultural threat).

The four categories represent distinct problem types requiring different interventions. Infrastructure deficits (PR1) call for community provision; colonial knowledge domination (PR2) calls for decolonial approaches; cultural erasure (PR3) calls for preservation; algorithmic opacity (PR4) calls for transparency. How problems are framed shapes what solutions become possible.

Condition categories

Infrastructure and Access Deficits (PR1) identifies lack of infrastructure or access as the core problem. This indicates absence or enclosure, and requires community provision. Guifi.net responds to broadband absence in rural areas where commercial ISPs find deployment unviable; Linux Foundation responds to proprietary software enclosure. The

mechanism is community / commons-based provision of what commercial actors will not provide or have enclosed.

Representative Cases: Guifi.net, Linux Foundation.

Colonial and Western-Centric Knowledge Domination (PR2) identifies dominance of Western / colonial knowledge systems as the core problem; this is epistemological marginalization requiring decolonial responses. Masakhane responds to Western dominance in AI development excluding African languages; Te Mana Raraunga responds to Western-centric data governance excluding Indigenous epistemologies.

Representative Cases: Masakhane, Te Mana Raraunga.

Cultural Erasure and Heritage Threat (PR3) identifies existential threats to cultural and linguistic survival as the core problem. The threat of erasure requires preservation. Monlam AI responds to political suppression of Tibetan language and cultural erasure through technological exclusion; Te Hiku Media responds to economic value frameworks ignoring cultural dimensions.

Representative Cases: Monlam AI, Te Hiku Media.

Algorithmic Opacity and Asymmetric Power (PR4) is about information asymmetry and lack of accountability in technological systems; opacity enables exploitation. Matrix responds to tensions between local control and global interoperability; Worker Algorithm Observatory responds to information asymmetry and algorithmic opacity in platform work. The mechanism is through strategies for transparency and community / worker-led accountability.

Representative Cases: Matrix, Worker Algorithm Observatory.

Cross-Type connections

Problem conditions frame the challenges that all other conditions address. PR1 (infrastructure deficits) motivates TT1 (community-controlled infrastructure) and TP3 (multi-

scale architecture). PR2 (colonial knowledge domination) motivates PU1 (Indigenous epistemological centering) and TP5 (culturally-grounded development). PR3 (cultural erasure) motivates PU2 (cultural preservation purpose) and TT2 (culturally-grounded artifacts). PR4 (algorithmic opacity) motivates TP1 (democratic data authority) and GV5 (inclusive expertise integration). Problem framing is not neutral. How challenges are named and internalized shapes what responses become visible.

Study of Condition Type Characteristics and Pathways

Technology Practices: Different Pathways to Technological Self-Determination

The Technology Practices range produces three distinct pathways through which initiatives build technological sovereignty.

The Data Authority Pathway (TP1) concentrates almost entirely within Data Sovereignty organizations (FNIGC, MAIAM NAYRI WINGARA, WAO) and health data initiatives (MiData, OpenSAFELY). These cases share a common mechanism: restructuring governance authority so that data subjects (Indigenous peoples, citizens, workers, patients) control decisions about data that concerns them. DECODE and OpenSAFELY represent Participatory Governance organizations employing this pathway, but even here the focus remains on data: DECODE on citizen data in smart cities, OpenSAFELY on health records.

The Open Development Pathway (TP2) concentrates heavily in Peer Production (Linux Foundation, Matrix, OpenStreetMap, Sensorica, WikiHouse) with limited extension to Digital Cooperatives (Loomio) and Participatory Governance (DECODE). The mechanism differs fundamentally from TP1: rather than restructuring authority over existing data, these initiatives create technology through transparent, collaborative processes that prevent enclosure. Linux Foundation builds shared infrastructure no single entity controls; WikiHouse enables distributed housing innovation; Matrix creates communication protocols anyone can implement. The pathway builds sovereignty through openness, making

technology impossible to enclose by making it collectively produced and universally available.

The Multi-Scale Architecture Pathway (TP3) spans organization types but concentrates in Peer Production (ARDC, Guifi.net, Matrix, OpenAgriNet) and extends to Digital Cooperatives (SOLshare) and Participatory Governance (Open Mobility Foundation). These cases deliberately architect technology to operate across scales simultaneously. Guifi.net's federated governance enables local community adaptation while maintaining network-wide coordination; SOLshare's peer-to-peer energy trading connects isolated solar systems; Open Mobility Foundation's data standards enable municipal control while ensuring regional interoperability. The mechanism resolves what might otherwise be a zero-sum tension between local autonomy and systemic integration.

The single-case Category 5 (Culturally-Grounded Development) deserves particular attention. Masakhane navigates the tension between using established NLP frameworks and adapting to language-specific community needs. This represents a pathway the current sample barely touches: technology development rooted in specific cultural epistemologies rather than generic technical solutions. The sparse evidence prevents strong empirical claims, but the theoretical significance is substantial. This may represent an underexplored dimension of technological sovereignty.

Resources: From Individual Empowerment to Economic Transformation

The Resources range is a spectrum from interventions targeting individual capacity to those restructuring economic relations.

Community Capacity Development (RS1) concentrates in Data Sovereignty organizations (DAIR, FNIGC, Masakhane) and Participatory Governance (Participatory Brazil, SDI, Stop LAPD Spying). The mechanism involves building collective capabilities for self-governance, training community researchers (FNIGC), developing local AI capacity

(Masakhane), organizing residents for advocacy (Stop LAPD Spying). All six cases share a common challenge: communities marginalized from technical and governance capacity. The intervention creates preconditions for other sovereignty dimensions.

Economic Value Redistribution (RS3) concentrates almost exclusively in Digital Cooperatives (CoopCycle, Fairbnb.coop, Namma Yatri, The Drivers Cooperative) with extension to Peer Production (Guifi.net, Solar Commons). The mechanism restructures value flows so surplus returns to workers, members, and communities rather than extracting to distant shareholders. Platform cooperatives (CoopCycle, The Drivers Cooperative, Fairbnb.coop, Namma Yatri) counter gig economy extraction by embedding drivers, riders, and hosts as owners. The concentration in cooperatives presents economic restructuring as the defining feature of cooperative identity.

The contrast between RS1 and RS3 illuminates different theories of change. RS1 builds capacity within existing economic structures, enabling communities to participate more effectively without necessarily transforming underlying relations. RS3 transforms the structures themselves, creating alternative circuits of value that operate outside extractive logics. Both contribute to sovereignty, but through different mechanisms. Notably, the highest-scoring Data Sovereignty cases (FNIGC, GIDA, Masakhane) combine RS1 capacity building with other conditions rather than RS3 economic restructuring. Their pathway runs through capacity and purpose.

Governance: Structural Permanence versus Process Intervention

The Governance range exposes a distinction between conditions that embed democracy in organizational structure versus those that intervene in decision-making processes.

Ownership-Based Democratic Control (GV2) appears exclusively in Digital Cooperatives (CoopCycle, Groupmuse, MiData, Som Energia, The Drivers Cooperative).

The mechanism anchors democratic governance in ownership structure: members govern because they own. This creates structural permanence; democratic control cannot be revoked without dissolving the ownership relation. The cooperative form itself becomes the governance mechanism, eliminating the gap between economic and political participation that characterizes capitalist enterprise.

Decentralized Authority Distribution (GV1) spans organization types (DAIR, Detroit CTC in Data Sovereignty; Nea Guinea/Tzoumakers in Peer Production; Loomio in Digital Cooperative; Enspiral, Participatory Brazil in Participatory Governance). Unlike GV2, these cases distribute authority without necessarily anchoring it in ownership. Enspiral creates collaborative networks across legally separate entities; Participatory Brazil redistributes municipal decision-making to neighborhood assemblies; Loomio enables distributed decision-making within organizations of any ownership form. The mechanism is process-focused, so authority flows differently without necessarily transforming underlying organizational form.

Federated Multi-Scale Coordination (GV3) appears in Peer Production (Guifi.net, Matrix) and Digital Cooperatives (MiData). These cases face a specific governance challenge: coordinating across scales without sacrificing local autonomy or network coherence. Guifi.net resolves this through governance architecture that preserves local community decision-making within network-wide coordination structures. Matrix enables federated server governance where each deployment maintains autonomy while participating in shared protocols. MiData coordinates across data cooperatives in different countries through federated structures.

Purpose: Reorientation versus Specific Commitments

The Purpose range denotes a hierarchy of foundational depth.

Indigenous Epistemological Centering (PU1) appears exclusively in Data Sovereignty organizations (FNIGC, GIDA, Te Hiku Media, Te Mana Raraunga). The mechanism goes beyond advocating for Indigenous rights to grounding governance in Indigenous worldviews. Good examples are the OCAP principles (Ownership, Control, Access, Possession) that guide FNIGC, the Māori Data Sovereignty principles that guide Te Mana Raraunga, and the cultural protocols embedded in Te Hiku Media's language technology. What is meant by reorientation is the centering of Indigenous epistemologies as the frameworks themselves.

Cultural Preservation (PU2) overlaps with PU1 in Data Sovereignty (Masakhane, Monlam AI, Te Hiku Media) but focuses specifically on linguistic and cultural preservation. Masakhane preserves African languages through NLP development; Monlam AI preserves Tibetan through AI tools; Te Hiku Media preserves te reo Māori through transcription and archiving. The mechanism treats technology as a means of cultural continuity, creating tools that strengthen linguistic and cultural diversity.

Value Reconceptualization (PU3) appears across organization types (Groupmuse in Digital Cooperative; MAIAM NAYRI WINGARA in Data Sovereignty; Solar Commons in Peer Production). These cases challenge dominant definitions of value: Groupmuse reconceptualizes house concerts as community events; Solar Commons frames energy as commons rather than commodity; MAIAM NAYRI WINGARA reframes Indigenous Australians as agents rather than problems. The mechanism operates at the level of meaning, contesting extractive value definitions rather than directly redistributing value.

Politics: Transformation versus Navigation

The Politics range presents the starkest contrast between transformative and adaptive engagement.

Policy and Systemic Transformation Advocacy (PO1) has the highest resonance of all 56 categories (41 connections) and appears across all organization types. Cases employing PO1 share a common mechanism: directly engaging political systems to reshape the environment in which community-led alternatives operate. FNIGC's advocacy produced the First Nations Data Governance Strategy; Te Mana Raraunga's advocacy influenced Statistics New Zealand's data policies; The Drivers Cooperative's advocacy shapes platform labor regulation.

Strategic Institutional Navigation (PO3) appears in only two cases (FNIGC, Namma Yatri) and receives a negative score (-0.25). The mechanism involves working within existing institutional frameworks. FNIGC navigates between federal and provincial jurisdiction; Namma Yatri navigates between platform regulation and cooperative law. The navigation may be strategically necessary (you cannot transform systems without surviving within them.) Notably, FNIGC combines PO3 navigation with PO1 advocacy, suggesting that effective political engagement often requires both survival strategies and transformation efforts.